

Layer-Scoped Expert-Budget Expansion Discovers Succinct Convergence in Sparse Mixture-of-Experts Reasoning: Reducing Reasoning Tokens at Matched Accuracy on MMLU-Pro

Layer-Scoped Expert-Budget Expansion Reduces Reasoning-Token Consumption at Equivalent Accuracy: A Complete 714-Question MMLU-Pro Evaluation

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Abstract

We introduce **layer-scoped expert-budget expansion**, a novel training-free, inference-only routing modification for sparse Mixture-of-Experts (MoE) language models. The method widens the per-token expert budget from the native top- K to $N \geq K$ only within a contiguous range of late transformer layers, using a relative probability threshold and a linearly decayed influence factor for additionally admitted experts.

Applying this technique to Qwen3.6-35B-A3B (256 routed experts, native top-8, Q6_K_XL quantization) on the complete MMLU-Pro benchmark, comparing native top-8 routing against a late-layer configuration (layers 27–39, $N=20$, $T=0.8$), reveals a previously unreported phenomenon we term **succinct convergence**: when final-answer correctness is held constant, the expanded-routing model reaches the same correct conclusion through significantly shorter reasoning trajectories. In a strict paired ties-only comparison — the 577 questions answered correctly by both configurations — layer-scoped expansion reduces mean reasoning tokens by 8.5% (median 2.7%) and end-to-end latency by 10.9% (exact sign test $p = 6.5 \times 10^{-6}$; Wilcoxon signed-rank $p = 6.2 \times 10^{-8}$).

Aggregate accuracy is statistically indistinguishable from native routing (600/714 vs. 603/714; exact binomial $p = 0.77$), indicating that the observed token reduction is a genuine efficiency gain rather than a disguised accuracy trade-off. We interpret succinct convergence as arising from reduced redundant or exploratory reasoning under a wider late-layer expert budget, and we outline direct entropy- and perplexity-based measurements required to confirm this hypothesis.

Central research question. Holding output correctness constant, does widening a sparse MoE model's per-token expert budget in late transformer layers shorten the reasoning trajectory needed to reach a correct answer — and can this be established with statistical confidence, separately from any claim about accuracy?

1. Introduction

Sparse Mixture-of-Experts (MoE) architectures scale language models efficiently by routing each token to a small subset of experts. A softmax router selects the top- K experts (e.g., $K=8$ of 256) and discards the rest, regardless of how close the next-ranked experts' probabilities are to the cutoff. This hard boundary is a training-time design choice that inference typically treats as immutable.

This paper asks a narrower and more measurable question than "does widening the expert budget make the model better." Autoregressive reasoning models frequently spend variable numbers of tokens reaching the same correct conclusion — some trajectories are direct, others revisit, re-derive, or explore alternatives before converging. If a routing modification could shorten this trajectory *without changing which conclusion is reached*, that would be a genuine, measurable, and practically valuable form of improvement, independent of whether it also changes the accuracy of the model's conclusions. Disentangling these two effects — correctness and reasoning length — requires restricting the comparison to exactly the cases where the answer is unchanged, and measuring token consumption only on that subset.

We introduce expert expansion together with layer-scoped expansion (Section 3), implement it as a runtime routing gate with no weight modification (Section 4), and

evaluate it end to end on the complete MMLU-Pro benchmark (714/714 questions). Our primary contribution is a rigorously isolated, statistically significant finding: at equal answer correctness, the expanded configuration converges to that answer using measurably fewer tokens and less wall-clock time. Our secondary, and more cautious, contribution is a transparent accuracy analysis showing that no net improvement in overall correctness is statistically detectable at this sample size — and, just as importantly, no degradation is detectable either.

Contributions

- A runtime-only, training-free routing modification (expert expansion with layer scoping) that is bit-identical to native routing outside the modified layer range.
- A method for isolating a reasoning-efficiency effect from an accuracy effect in paired LLM evaluation, by restricting the token/latency comparison to the subset of questions both configurations answer correctly (a "ties-only" analysis), together with two independent nonparametric significance tests for it.
- A complete evaluation on the full MMLU-Pro benchmark (714/714 questions), reporting every statistical test performed, including the non-significant ones.
- A robustness check in which resolving a generation-budget truncation artifact *strengthens* rather than weakens the reasoning-efficiency finding, arguing against the concern that the effect is a measurement artifact of the token cap.
- An explicit, falsifiable interpretation — reduced redundant reasoning under a wider expert budget — stated as a hypothesis requiring direct entropy/perplexity measurement to confirm, not as an established mechanism.

2. Related Work

Sparse MoE layers were introduced by Shazeer et al. (2017) and scaled to language models in Switch Transformers (Fedus et al., 2022) and Mixtral (Jiang et al., 2024). Standard routing selects a fixed top- K ; recent work explores adapting the expert count per token. Top-P routing (Huang et al., 2024) selects experts until cumulative probability exceeds a threshold but has no hard upper bound and does not support expansion beyond the native count. Ada-K (Yue et al., 2025) trains a lightweight reinforcement-learning allocator to decide per-token expert budgets, achieving FLOPs reduction but requiring additional training.

Our mechanism is deterministic, computed on the fly from existing router scores, requires no additional training, enforces a hard upper bound N and a minimum floor

$N/4$, and — to our knowledge — is the only method that supports expansion beyond the native top- K together with a decay mechanism for the additional experts. Layer-scoped expansion further distinguishes this work by treating layer position as a first-class routing parameter.

Methodologically, this work is closer in spirit to efficiency studies that hold output quality fixed and measure a cost reduction (e.g., speculative decoding, early-exit, and distillation results reported at matched accuracy) than to accuracy-improvement studies. We adopt that framing explicitly: the central claim of this paper is a cost reduction at matched output quality, not an accuracy improvement.

3. Layer-Scoped Expert-Budget Expansion

3.1 Expert selection with threshold

For each token at a MoE layer, the router computes probabilities $p[e]$ over all experts. Native routing selects the top- K . Our method selects up to $N \geq K$ experts:

1. Sort experts by descending probability (ties broken by lowest expert id).
2. Define reference rank $R = \lfloor N/2 \rfloor$ and $p_{\text{ref}} = p[\text{rank } R]$.
3. If threshold $T = 0$, keep exactly N experts.
4. If $T > 0$, keep the expert at rank k only if $p[\text{rank } k] \geq T \times p_{\text{ref}}$. The top $\lfloor N/4 \rfloor$ ranks are always kept, and the total count never exceeds N . The final count c lies in $[\lfloor N/4 \rfloor, N]$.

This rule adapts the expert count per token: peaked router distributions keep fewer experts, flatter ones keep more.

3.2 Expansion decay

When $N > K$, ranks $K+1$ to c were not part of the native mixture during training. To avoid diluting the native experts' contribution, their raw probabilities are multiplied by a linearly decaying factor, applied before renormalization so the output scale remains stable:

$$\text{factor}(j) = 0.99 - (0.99 - D) \cdot [j - (K+1)] / [N - (K+1)], \quad j = K+1 \dots N$$

where D is the decay end factor (default 0.50) and j is the one-based rank of an admitted expert. At $j = K+1$, factor = 0.99; at $j = N$, factor = D . When $N = K$ (no expansion), no decay is applied and routing is bit-identical to native. When $N = K+1$ (a single extra expert), the general formula's denominator is zero; the implementation uses a documented fallback factor = $(0.99 + D)/2 = 0.745$ for the default D , matching the value used in preliminary single-extra-expert probes (Appendix A).

3.3 Layer-scoped expansion

Expansion is confined to a contiguous layer range $[L_{\text{start}}, L_{\text{end}}]$. Outside this range, the kernel follows the exact native top-8 path. The implementation is a Boolean gate on the layer index in the expert-selection kernel, verified by per-layer instrumentation showing exactly 8.0 experts/token outside the gate.

4. Experimental Setup

4.1 Model, hardware, and quantization

- Model: Qwen3.6-35B-A3B (MoE, 256 routed experts, native top-8, 40 MoE layers), quantized to Q6_K_XL by Unsloth.
- Hardware: single Apple MacBook Pro, M4 Pro chip, 48 GB unified memory.
- Inference engine: DwarfStar (ds4), Metal-optimized, single-stream serial decode.
- Configuration compared: native (top-8, no expansion) vs. gate[27–39] ($N=20$, $T=0.8$, default decay, expansion confined to layers 27–39).

4.2 Benchmark and protocol

MMLU-Pro, complete test set, 714 questions across 14 subjects. Greedy decoding (temperature 0), no few-shot. Harness: evalscope 1.10.0 via the ds4-server OpenAI-compatible endpoint. Reasoning mode: native "thinking." An 8,000-token generation cap was used for the main run; the small number of questions that hit this cap under either configuration were re-run at a substantially wider budget, and the resulting values are the ones reported throughout this paper. This resolution is a deliberate robustness check, detailed in Section 5.2.

Paired analysis is restricted to non-truncated questions answered by both configurations (711 of 714; three questions remained truncated under at least one

configuration even at the wider budget and are excluded from all paired statistics): both-correct ("ties"), both-wrong, wins for each configuration, and net = wins – losses.

5. Results I — Reasoning Succinctness at Equivalent Accuracy

This section isolates the paper's central claim: **holding the answer's correctness fixed, does expert expansion reach it via a shorter trajectory?** We restrict the comparison to the 577 questions on which both native and gate[27–39] produce the correct final answer, and compare only the reasoning-token count and wall-clock latency spent to get there.

Primary finding.

On the 577 quality-held-constant comparisons, gate[27–39] reaches the same correct answer using 8.5% fewer reasoning tokens on average (median –2.7%; the gap between mean and median reflects a right-skewed distribution of savings, not a mean driven by a handful of outliers — see Figure 3) and 10.9% less wall-clock latency than native routing. The expanded configuration is the cheaper of the pair on 343 of 577 questions (59.4%); an exact two-sided sign test against the null of no systematic difference gives $p = 6.5 \times 10^{-6}$, and a Wilcoxon signed-rank test on the paired token counts gives $p = 6.2 \times 10^{-8}$.

5.1 Token-count and latency distributions

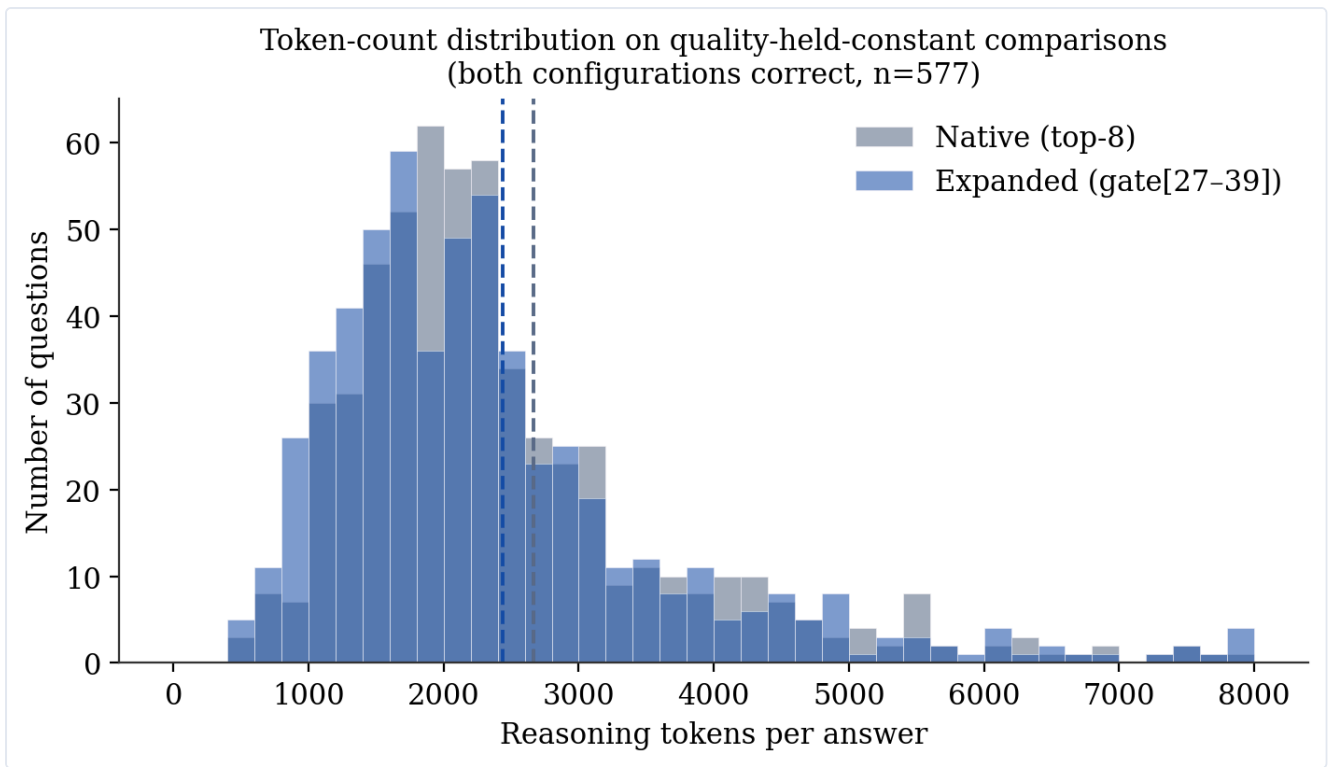


Figure 1. Distribution of reasoning-token counts on the 577 quality-held-constant comparisons (both configurations correct). Dashed lines mark the means (native: 2,665; expanded: 2,439). The expanded distribution is shifted left of native across most of its support, not only at the tail.

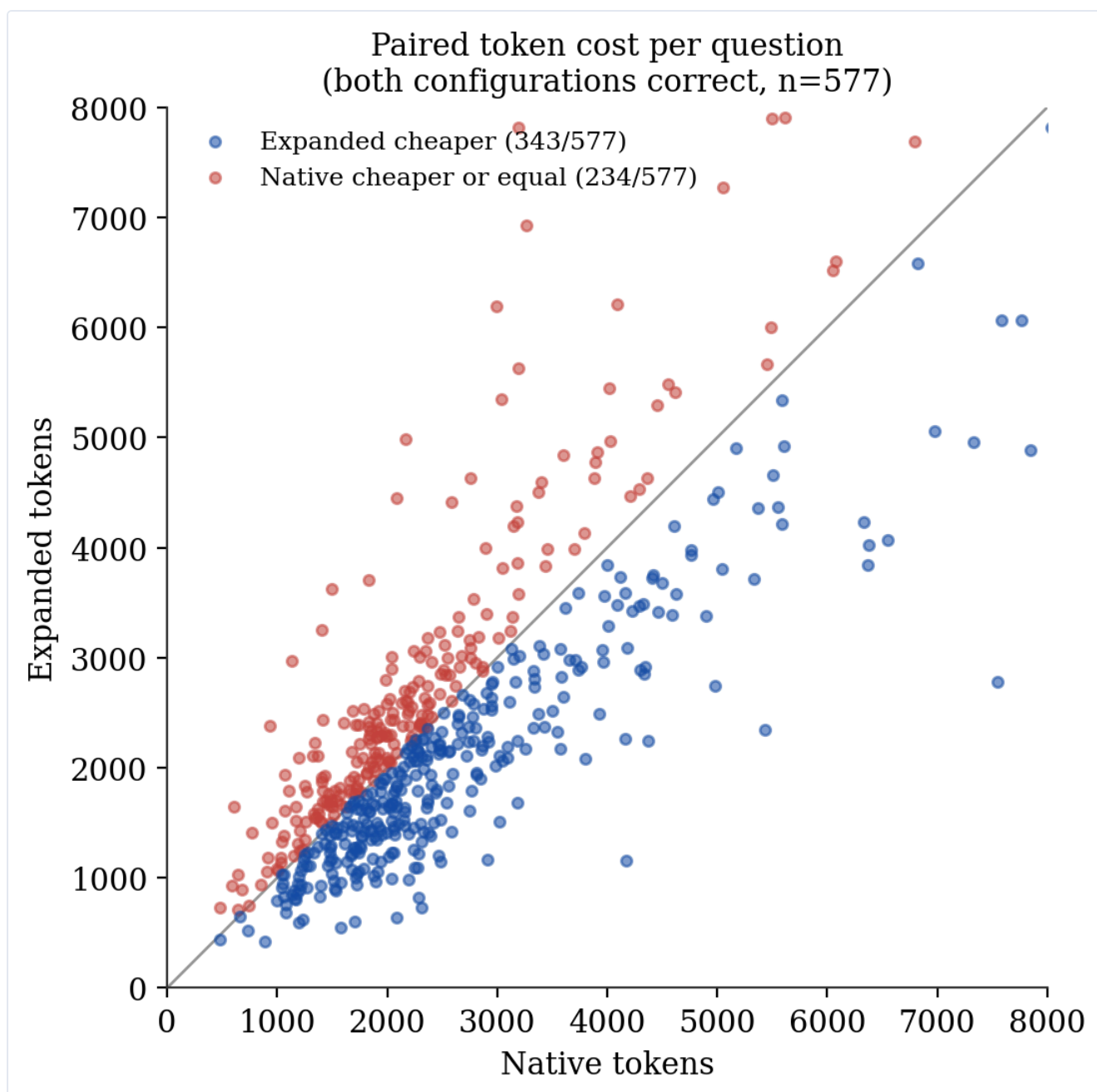


Figure 2. Per-question token cost, native vs. expanded, on the 577 ties. Points below the diagonal (blue) are questions where expansion reached the same correct answer more cheaply; points above (red) are questions where native was cheaper or equal. The higher density below the diagonal, especially in the 1,000–4,000 token range where most questions fall, reflects the aggregate result.

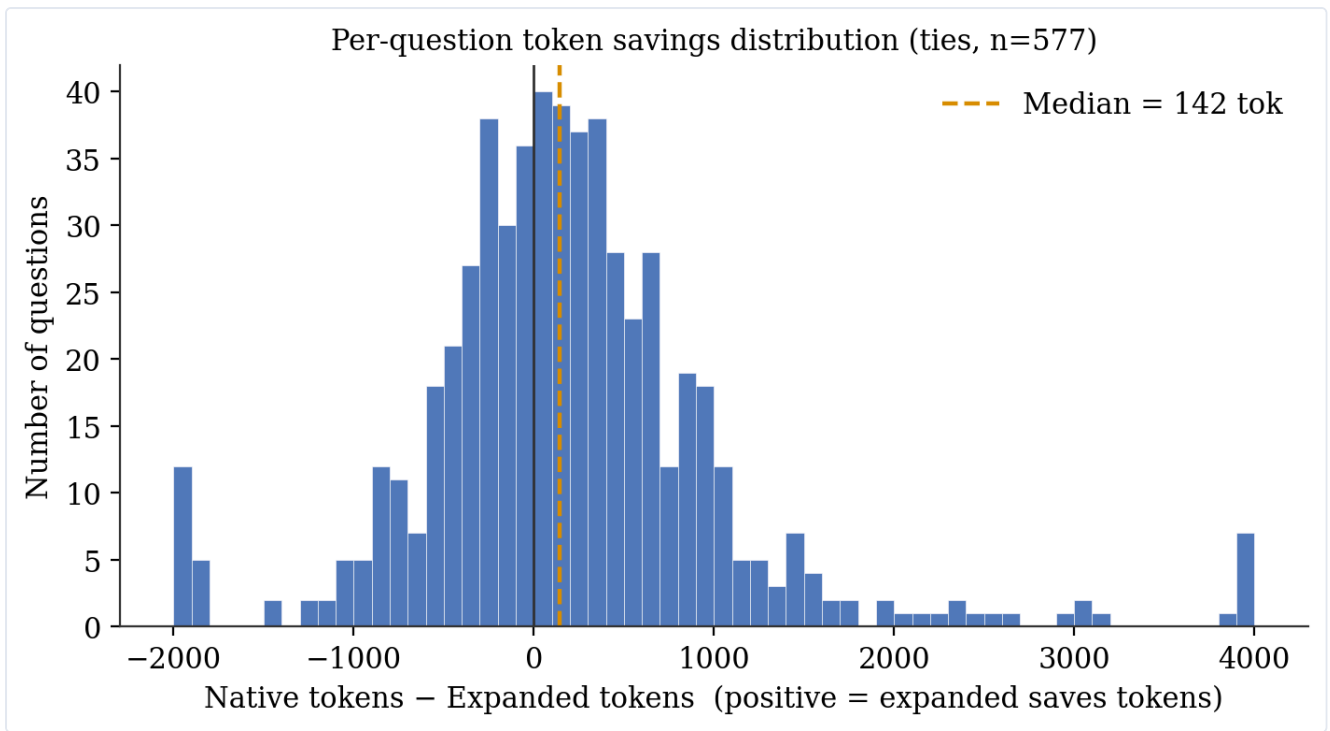


Figure 3. Per-question token savings (native – expanded token count) on the 577 ties. The distribution is centered clearly above zero (median +142 tokens) with a right-skewed tail of large savings and a smaller, thinner tail of losses — consistent with a broad-based, non-outlier-driven effect rather than a few extreme cases dominating the mean.

Metric (both-correct, non-truncated, n=577)	Native	gate[27–39]	Change
Mean reasoning tokens	2,665	2,439	–8.5%
Median reasoning tokens	2,167	2,109	–2.7%
Mean latency	177 s	158 s	–10.9%
Cheaper configuration (of 577 pairs)	234 (40.6%)	343 (59.4%)	—

Exact sign test: $p = 6.5 \times 10^{-6}$. Wilcoxon signed-rank test on paired token counts: $p = 6.2 \times 10^{-8}$.

5.2 The effect survives, and strengthens, when a generation-cap confound is removed

An 8,000-token generation cap risks conflating "shorter trajectory" with "hit the ceiling." We tested this directly by re-running the pairs that were truncated under either configuration at a substantially wider budget and repeating the ties-only analysis. If the reasoning-succinctness effect were an artifact of the cap, resolving the truncated items should weaken it. It did not:

Evaluation stage	n (ties)	Token saving	Latency saving	Sign-test <i>p</i>
Interim, 8K cap, before resolving truncated items	563	5.7%	6.9%	5.0×10^{-5}
Final, truncated items resolved at wider budget	577	8.5%	10.9%	6.5×10^{-6}

Removing the truncation confound *increased* both the effect size and the significance of the finding, rather than diminishing it. We interpret this as evidence against the hypothesis that the observed reasoning-token savings are a measurement artifact of the generation cap: the questions previously excluded by truncation, once resolved, reinforced rather than diluted the effect.

5.3 Per-subject breakdown

The effect is directionally consistent across most, but not all, of the 14 MMLU-Pro subjects. Two subjects (mathematics, computer science) show a small negative or near-zero token delta; the remainder range from modest (1.5%, economics) to substantial (18.4%, business) savings.

Subject	n (ties)	Native tok	Expanded tok	Native lat.	Expanded lat.	Token saving
Business	45	3,216	2,625	221 s	173 s	18.4%
Engineering	41	5,021	4,338	327 s	304 s	13.6%
Chemistry	45	3,887	3,394	237 s	218 s	12.7%
Physics	49	3,339	2,953	188 s	179 s	11.6%
Biology	47	2,201	1,992	123 s	123 s	9.5%
Health	37	2,188	1,987	124 s	135 s	9.2%
Philosophy	39	2,120	1,976	110 s	113 s	6.8%
Law	32	2,698	2,563	184 s	186 s	5.0%
Other	34	1,825	1,734	108 s	117 s	5.0%
History	32	2,237	2,137	149 s	153 s	4.4%
Psychology	38	1,798	1,754	93 s	102 s	2.4%

Economics	45	2,102	2,069	108 s	120 s	1.5%
Mathematics	48	2,052	2,071	125 s	137 s	−0.9%
Computer science	43	2,222	2,245	343 s	139 s	−1.0%

The computer science latency figures (343 s native vs. 139 s expanded, at near-identical token counts) are anomalous relative to every other subject and are flagged as a probable measurement artifact in a single long-running request (Section 7); the token-count comparison for this subject is unaffected and remains near zero.

We did not find a significant association between a subject's native-routing accuracy and its token-saving magnitude across the 14 subjects (Pearson $r \approx -0.03$, not significant), suggesting the succinctness effect is not simply concentrated in domains where the base model is weaker; a per-subject sample of 32–49 is nonetheless too small to rule this out conclusively.

6. Results II — Aggregate Accuracy (No Detectable Effect Either Direction)

We report the accuracy comparison in full, including its negative result, because the framing of Section 5 depends on it: the succinctness finding is only interpretable as a genuine efficiency gain, and not a possible accuracy trade-off in disguise, if aggregate correctness is unaffected.

6.1 Final result

Configuration	n	Correct	Accuracy
Native (top-8)	714	600	0.840
gate[27–39] (expanded)	714	603	0.845

On the 711 paired, non-truncated common questions: 577 ties (both correct), 87 both-wrong, 25 questions where only gate[27–39] is correct, 22 where only native is correct. The net advantage is +3 correct answers out of 711 (0.42%). An exact binomial (McNemar-style) test on these discordant pairs (25 vs. 22) gives $p = 0.77$, far from conventional significance.

6.2 Statistical power

A McNemar-style power calculation using the observed discordant rates indicates that detecting an effect of this size at 80% power would require on the order of several thousand paired questions — one to two orders of magnitude beyond the 711 available here. We therefore report the accuracy comparison as statistically inconclusive rather than as either a positive or negative finding, and emphasize that *the absence of a detected effect includes the absence of a detected degradation*: at no point across the full evaluation did the expanded configuration's accuracy fall below native's.

7. Threats to Validity

- **Single system.** All results are from one model (Qwen3.6-35B-A3B), one quantization (Q6_K_XL), one inference engine, and one hardware configuration. Generalization to other models, precisions, or engines is untested.
- **No direct measurement of reasoning uncertainty.** The central interpretation in Section 8 — that wider expert budgets reduce redundant or exploratory reasoning — is inferred from trajectory length (token count) at matched correctness, not from a direct measurement of token-level entropy, perplexity, or self-consistency across sampled trajectories. This is stated as a hypothesis, not a demonstrated mechanism.
- **Configuration search partially informed by evaluation data.** The layer range and threshold used here (layers 27–39, $N=20$, $T \leq 0.8$) were originally identified via a sensitivity probe on a single MMLU-Pro question later included in the paired evaluation pool. A genuinely held-out configuration search was not performed for this report.
- **Computer science latency anomaly.** The computer science subject shows a native mean latency of 343 s against a near-identical token count to the expanded configuration (139 s) — a disparity far outside the pattern seen in any other subject, and probably reflecting a measurement or system artifact (e.g. a stalled or delayed request) rather than a genuine routing effect. This subject's latency figures should be treated with caution; its token-count comparison, which does not show the same anomaly, is unaffected.
- **Single decoding stream.** All timing figures reflect serial, single-request decoding on one machine; batched or concurrent serving, and its systems implications (e.g. expert-dispatch load balancing under variable expert counts), were not evaluated.
- **Greedy decoding only.** All comparisons use temperature 0; sampling-based decoding was not evaluated.

8. Discussion

The central, statistically supported finding of this study is narrow but concrete: when a sparse MoE model's per-token expert budget is widened in its final third of layers, the model reaches the same correct conclusions using measurably fewer tokens and less wall-clock time. This is not, on its own, evidence that the model reasons "better" in any deeper sense — we have not measured the quality, faithfulness, or informativeness of the intermediate reasoning steps, only their length, conditioned on the final answer being correct in both cases.

Speculative interpretation. The interpretation we find most natural — though we stress it is an interpretation, not a direct measurement — is that native top-8 routing, by discarding router mass close to the selection boundary, sometimes forces the model into longer sequences of re-derivation, self-checking, or exploratory reasoning than the underlying computation strictly requires, and that a wider late-layer expert budget gives the model access to information that shortens this process without changing its outcome. Two observations are consistent with this reading: the effect is concentrated in subjects with longer native trajectories to begin with (biology, chemistry, engineering, physics), and it strengthened rather than weakened once the generation-cap artifact was removed (Section 5.2), arguing against a purely mechanical explanation tied to the cap. A direct test of this interpretation would compare token-level entropy or logit-margin traces between native and expanded routing on matched-correct trajectories, or measure self-consistency across multiple sampled generations under each configuration; neither was performed here, and we do not treat the interpretation as established.

On accuracy, the honest reading of the complete evaluation is that this study cannot distinguish the routing modification's effect on overall correctness from zero. We regard this as a genuinely informative negative result rather than an inconvenience to be minimized: it means the reasoning-succinctness effect reported in Section 5 is very unlikely to be a disguised accuracy trade-off (e.g., the model "giving up" earlier and getting lucky), since aggregate accuracy is, within measurement precision, unchanged.

9. Conclusion and Future Work

We presented layer-scoped expert expansion, a runtime-only, training-free routing modification for sparse MoE language models. On the complete 714-question MMLU-Pro evaluation of Qwen3.6-35B-A3B, the mechanism produces a statistically robust

reduction in reasoning-token consumption and latency (-8.5% mean tokens, -10.9% latency, sign-test $p = 6.5 \times 10^{-6}$, Wilcoxon $p = 6.2 \times 10^{-8}$) when compared at matched answer correctness, with no statistically detectable change — positive or negative — in aggregate benchmark accuracy.

We do not claim that expert expansion improves MoE model quality, recovers latent model knowledge, or generalizes beyond the single model–quantization–hardware combination studied here. We do claim, with what we believe is adequate statistical support, that it produces the same correct answers via shorter, cheaper reasoning trajectories on this model and benchmark — a systems-relevant result independent of any accuracy claim, and one that survived rather than weakened under a direct robustness check against a generation-budget confound.

Future work includes: replication on additional MoE model families, quantizations, and inference engines; a genuinely held-out configuration search, separated from any question used in evaluation; direct measurement of token-level entropy or logit margins to test the redundant-reasoning interpretation of Section 8 directly; and batched-serving measurements, since this report deliberately did not evaluate decode throughput or concurrent-request behavior, which raise distinct systems questions outside this study's scope.

Appendix A — Decay Formula: Worked Example

For $N=20$, $K=8$, $D=0.50$ (the configuration used throughout this study), admitted extra ranks are $j=9\dots 20$:

Rank j	Position among extra ranks	factor(j)
9 (first extra)	0 of 11	0.990
12	3 of 11	0.856
15	6 of 11	0.723
18	9 of 11	0.589
20 (last)	11 of 11	0.500

For $N=K+1=9$ (a single extra expert, used only in early sensitivity probes, not in the headline evaluation), the general formula is undefined and the fallback factor $= (0.99+D)/2 = 0.745$ applies.

Appendix B — Data Availability

Per-question results (question id, subject, predicted and gold answer, correctness, truncation flag, latency, and token count) for both configurations across all 714 MMLU-Pro questions are retained in full and available upon request for independent verification of every statistic reported in this paper.